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Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. I am an ML engineer with more than five years of experience in applied machine learning and NLP, including taking document models from development into production. I hold an MSc in Computer Science and am currently pursuing a PhD focused on named entity recognition in Croatian legal texts.

At Slavonica Document AI, I developed and deployed an incoming-document classifier for public administration that achieves 94% accuracy and a 0.91 macro F1 across 23 classes, processing approximately 15,000 documents per day. I fine-tuned XLM-R and BERTić models and delivered them as FastAPI services in Docker on Azure App Service, using Azure ML and MLflow. I used MLflow for experiment tracking and model registry, alongside strong Python-based development for production model serving. I also established weekly production monitoring; it identified an OCR change that reduced accuracy by nine percentage points, allowing the issue to be resolved within a week.

My earlier work at Baranjska kreditna banka is directly relevant to customer support: I built an 11-topic classifier for customer complaints, progressing from a TF-IDF and logistic-regression baseline to fastText. Its outputs supported weekly reporting to customer service management. I have also built a weekend prototype for question answering over Croatian laws using sentence embeddings, FAISS and an open-weight LLM accessed through an API. Although this was a personal prototype rather than a production system, it gave me hands-on experience with retrieval-augmented question answering using sentence embeddings, FAISS and an open-weight LLM accessed through an API.

I am keen to bring my experience in NLP, Python-based model serving, production monitoring and MLOps with Azure ML and MLflow to Quillstone's AI team as it builds tools to answer merchant questions and support customer operations. The team's emphasis on ownership, practical problem-solving and learning together particularly appeals to me. I would welcome the opportunity to discuss how my background could contribute to the role.

Kind regards,

Petra Zlatarić-Novosel